Cloud Lab - 9

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1) Installing Helm Package Manager

```
/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ curl -fsSL -o get_helm.sh https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-4
  chmod 700 get_helm.sh
  ./get_helm.sh
Downloading https://get.helm.sh/helm-v4.1.3-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
[sudo] password for harisshragav:
helm installed into /usr/local/bin/helm
```

Adding OpenFAAS Helm Repo

```
/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁  took 15s
❯ helm repo add openfaas https://openfaas.github.io/faas-netes/

"openfaas" has been added to your repositories

/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "openfaas" chart repository
Update Complete. ⎈Happy Helming!⎈
```

Creating Auth Password

```
/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ kubectl -n openfaas create secret generic basic-auth \
          --from-literal=basic-auth-user=admin \
          --from-literal=basic-auth-password=abc12
secret/basic-auth created
```

Installing OpenFaaS

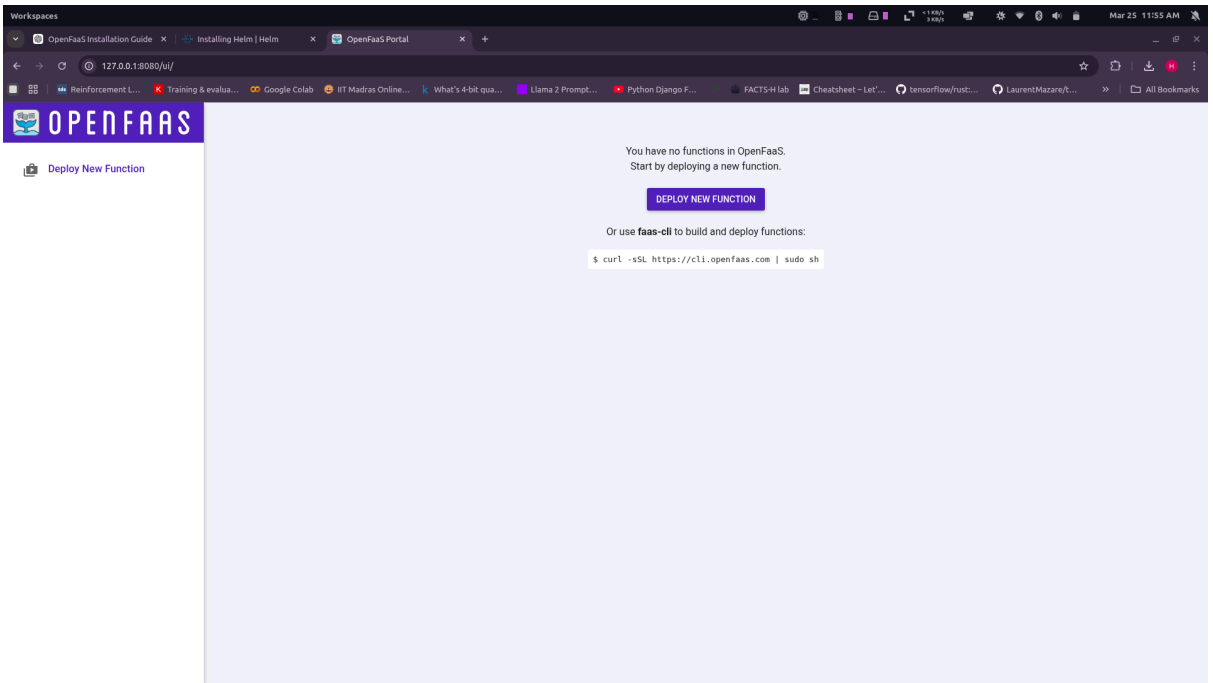
```
/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ helm upgrade openfaas openfaas/openfaas \
        --install \
        --namespace openfaas \
        --set functionNamespace=openfaas-fn \
        --set generateBasicAuth=true
Release "openfaas" does not exist. Installing it now.
NAME: openfaas
LAST DEPLOYED: Wed Mar 25 11:43:06 2026
NAMESPACE: openfaas
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
TEST SUITE: None
NOTES:
To verify that openfaas has started, run:

  kubectl -n openfaas get deployments -l "release=openfaas, app=openfaas"

To retrieve the admin password, run:

  echo $(kubectl -n openfaas get secret basic-auth -o jsonpath="{.data.basic-auth-password}" | base64 --decode)
```

Verifying Authentication and Installation

```
/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ echo (kubectl -n openfaas get secret basic-auth -o jsonpath="{.data.basic-auth-password}" | base64 --decode)
fyfc8EzARG5cvhtByixGeU5bt2Duk8No

/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ kubectl -n openfaas get deployments -l "release=openfaas, app=openfaas"
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
alertmanager   0/1     1            0           33s
gateway        0/1     1            0           33s
nats           0/1     1            0           33s
prometheus     0/1     1            0           33s
queue-worker   0/1     1            0           33s

/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯
```

Port forwarding OpenFAAS Gateway to setup its UI

```
/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ kubectl port-forward svc/gateway -n openfaas 8080:8080
Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
```



Logging in fast-cli

```
/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ faas-cli login --username admin --password fyfc8EzARG5cvhtByixGeU5bt2Duk8No
WARNING! Using --password is insecure, consider using: cat ~/faas_pass.txt | faas-cli login -u user --password-stdin
Calling the OpenFaaS server to validate the credentials...
credentials saved for admin http://127.0.0.1:8080

/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯
```

Pulling python3-http template for OpenFAAS

```
/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁
❯ faas-cli template store pull python3-http
2026/03/25 12:00:20 Fetching templates from https://github.com/openfaas/python-flask-template
Wrote 5 template(s) : [python27-flask python3-flask python3-flask-debian python3-http python3-http-debian]

/mnt/2028B41628B3E944/Projects/OpenFAAS on ☁  took 2s
❯
```

Creating a test fn

```
❯ faas-cli new test-fn --lang python3-http
Folder: test-fn created.
  ___                   _____       ____
 / _ \ _ __  ___ _ __ |_   _|_ _  __ ___ \
| | | | '_ \ / _ \ '_ \  | |/ _` |/ _` |__) |
| |_| | |_) |  __/ | | | | | (_| | (_| |__) |
 \___/| .__/ \___|_| |_|_|  \__,_|\__,_|____/
      |_|


Function created in folder: test-fn
Stack file written: stack.yaml
```

```
test-fn
  handler_test.py
  handler.py
  requirements.txt
  tox.ini
.gitignore
get_helm.sh
stack.yaml
```

handler.py

```
def handle(event, context):
    return {
        "statusCode": 200,
        "body": "Hello from OpenFaaS!"
    }
```

Using minikube docker, deploying the test-fn

stack.yaml

```
version: 1.0
provider:
 name: openfaas
 gateway: http://127.0.0.1:8080
functions:
 test-fn:
   lang: python3-http
   handler: ./test-fn
   image: gl3mon/test-fn:latest
```

```
/mnt/2028B41628B3E944/Projects/OpenFAAS via λ <custom snapshot> on ☁️  took 4s
❯ faas-cli up -f stack.yaml
[0] > Building test-fn.
Building: gl3mon/test-fn:latest with python3-http template. Please wait..
2026/03/25 12:09:25 Build flags: [build --build-arg ADDITIONAL_PACKAGE= --tag gl3mon/test-fn:latest .]
#0 building with "default" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 1.69kB done
#1 WARN: RedundantTargetPlatform: Setting platform to predefined ${TARGETPLATFORM:-linux/amd64} in FROM is redundant as this is the default behavior (line 3)
#1 WARN: RedundantTargetPlatform: Setting platform to predefined ${TARGETPLATFORM:-linux/amd64} in FROM is redundant as this is the default behavior (line 4)
#1 DONE 0.0s

#2 [auth] library/python:pull token for registry-1.docker.io
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.12-alpine
#3 ...

#4 [internal] load metadata for ghcr.io/openfaas/of-watchdog:0.11.4
#4 DONE 0.9s

#3 [internal] load metadata for docker.io/library/python:3.12-alpine
#3 DONE 2.6s

#5 [internal] load .dockerignore
#5 transferring context: 2B done
#5 DONE 0.0s

#6 [watchdog 1/1] FROM ghcr.io/openfaas/of-watchdog:0.11.4@sha256:a29144e771567383e6bdb06a82cd22575160f1bf358bb0a3ea3b38ffeae22da5
#6 DONE 0.0s

#7 [build  1/16] FROM docker.io/library/python:3.12-alpine@sha256:7747d47f92cfca63a6e2b50275e23dba8407c30d8ae929a88ddd49a5d3f2d331
#7 DONE 0.0s
```

```
/mnt/2028B41628B3E944/Projects/OpenFAAS via λ <custom snapshot> on ☁️  took 4s
❯ faas-cli up -f stack.yaml
#8 [internal] load build context
#8 transferring context: 3.38kB done
#8 DONE 0.0s

#9 [build  3/16] RUN chmod +x /usr/bin/fwatchdog
#9 CACHED

#10 [build 14/16] COPY --chown=app:app function/requirements.txt        .
#10 CACHED

#11 [build  6/16] RUN chown app /home/app
#11 CACHED

#12 [build  2/16] COPY --from=watchdog /fwatchdog /usr/bin/fwatchdog
#12 CACHED

#13 [build 13/16] WORKDIR /home/app/function/
#13 CACHED

#14 [build 16/16] COPY --chown=app:app function/   .
#14 CACHED

#15 [build  9/16] COPY --chown=app:app requirements.txt   .
#15 CACHED

#16 [build  5/16] RUN addgroup -S app && adduser app -S -G app
#16 CACHED

#17 [build 11/16] RUN mkdir -p function
#17 CACHED

#18 [build 12/16] RUN touch ./function/__init__.py
#18 CACHED

#19 [build  4/16] RUN if [ "${UPGRADE_PACKAGES}" = "true" ] || [ "${UPGRADE_PACKAGES}" = "1" ]; then apk --no-cache upgrade; fi &&     apk --no-cache add
#19 CACHED

#20 [build 10/16] RUN pip install --no-cache-dir -r requirements.txt
#20 CACHED

#21 [build 15/16] RUN pip install --no-cache-dir --user -r requirements.txt
#21 CACHED

#22 [build  7/16] WORKDIR /home/app/
#22 CACHED

#23 [build  8/16] COPY --chown=app:app index.py           .
#23 CACHED

#24 [ship 1/1] WORKDIR /home/app/
#24 CACHED
```

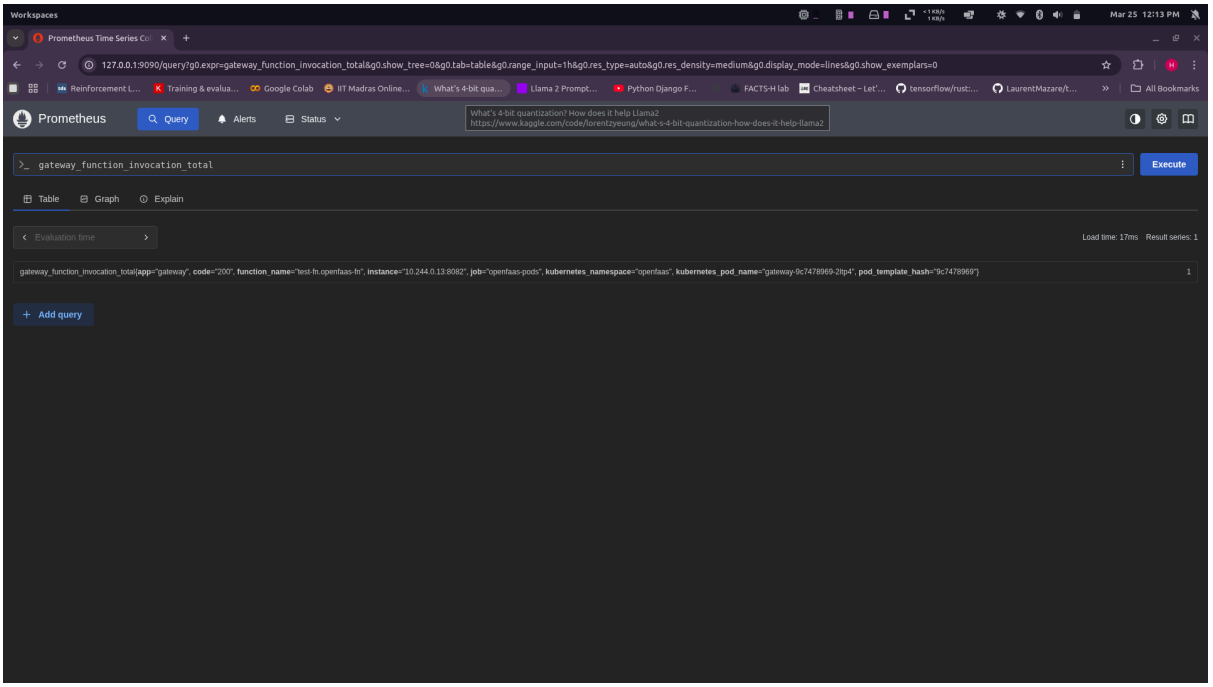
```
/mnt/2028B41628B3E944/Projects/OpenFAAS via λ <custom snapshot> on ☁  took 4s
❯ faas-cli up -f stack.yaml
#23 [build  8/16] COPY --chown=app:app index.py           .
#23 CACHED

#24 [ship 1/1] WORKDIR /home/app/
#24 CACHED

#25 exporting to image
#25 exporting layers done
#25 writing image sha256:bbd38f5bb555312d1d581fbd2b99948bc3ab0127f90c578dc9af82c12711e283 done
#25 naming to docker.io/gl3mon/test-fn:latest done
#25 DONE 0.0s

 2 warnings found (use docker --debug to expand):
 - RedundantTargetPlatform: Setting platform to predefined ${TARGETPLATFORM:-linux/amd64} in FROM is redundant as this is the default behavior (line 3)
 - RedundantTargetPlatform: Setting platform to predefined ${TARGETPLATFORM:-linux/amd64} in FROM is redundant as this is the default behavior (line 4)
Image: gl3mon/test-fn:latest built.
[0] < Building test-fn done in 2.97s.
[0] Worker done.

Total build time: 2.97s
[0] > Pushing test-fn [gl3mon/test-fn:latest]
The push refers to repository [docker.io/gl3mon/test-fn]
5f70bf18a086: Mounted from prom/alertmanager
f165e3a24a1a: Pushed
503821d34cc5: Pushed
f92520bf5c0f: Pushed
ce8bf13fa029: Pushed
0ff1750ae3d8: Pushed
e4de04e5ff40: Pushed
db7f641250ff: Pushed
82425fc23368: Pushed
04be5588f634: Pushed
6e1f6632b813: Pushed
3d873c1d5eb1: Pushed
281107ac692f: Pushed
7827ee2487d1: Pushed
f045c92e01b4: Mounted from library/python
60285744fe26: Mounted from library/python
2a320211b927: Mounted from library/python
989e799e6349: Mounted from library/python
latest: digest: sha256:e8e2d8c5e709194f70af22b0c9207b029f0f74cc814bf80abc99cb4e6638ef29 size: 4482
[0] < Pushing test-fn [gl3mon/test-fn:latest] done.
[0] Worker done.
Deploying: test-fn.

Deployed. 202 Accepted.
URL: http://127.0.0.1:8080/function/test-fn


/mnt/2028B41628B3E944/Projects/OpenFAAS via λ <custom snapshot> on ☁  took 41s
❯
Ln 11, Col 1 (175 selected)   Spaces: 2   UTF-8   LF
```

Testing the test function endpoint

```
/mnt/2028B41628B3E944/Projects/OpenFAAS via λ <custom snapshot> on ☁  took 41s
❯ curl http://127.0.0.1:8080/function/test-fn
Hello from OpenFaaS!
/mnt/2028B41628B3E944/Projects/OpenFAAS via λ <custom snapshot> on ☁
❯
Ln 11, Col 1 (175 selected)   Spaces: 2   UTF-8   LF
```

To monitor, port forward to prometheus

```
/mnt/2028B41628B3E944/Projects/OpenFAAS via λ <custom snapshot> on ☁
❯ kubectl port-forward svc/prometheus -n openfaas 9090:9090
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
```

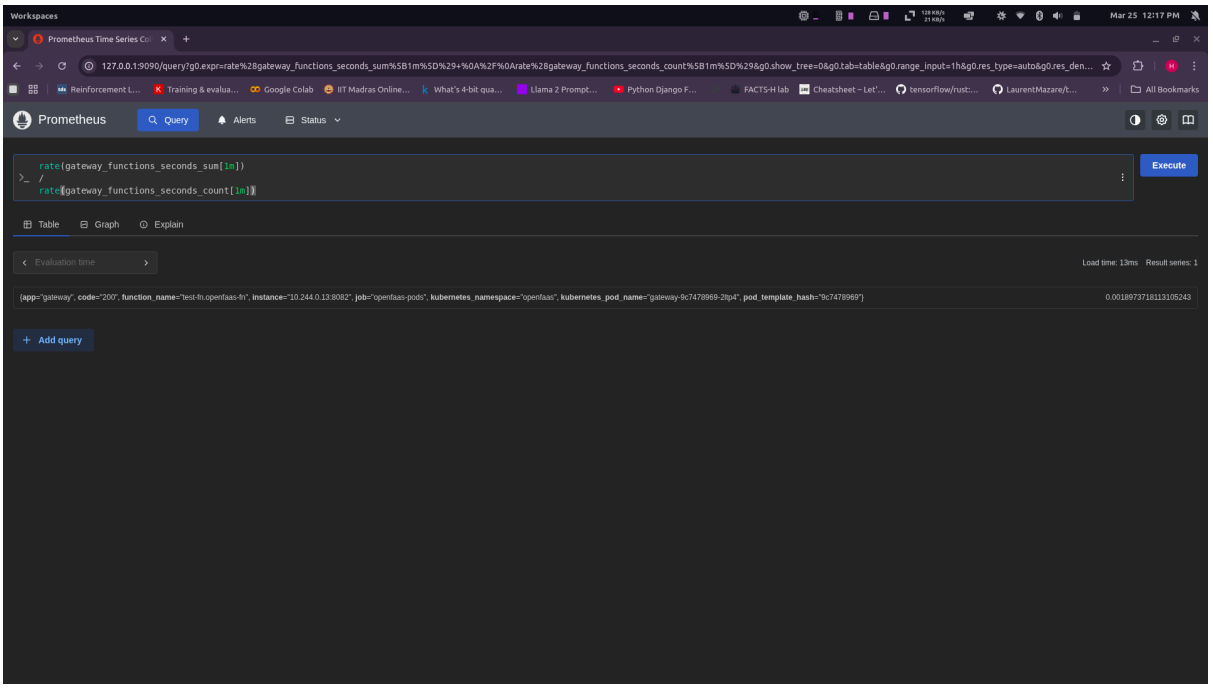


Creating traffic to monitor in prometheus

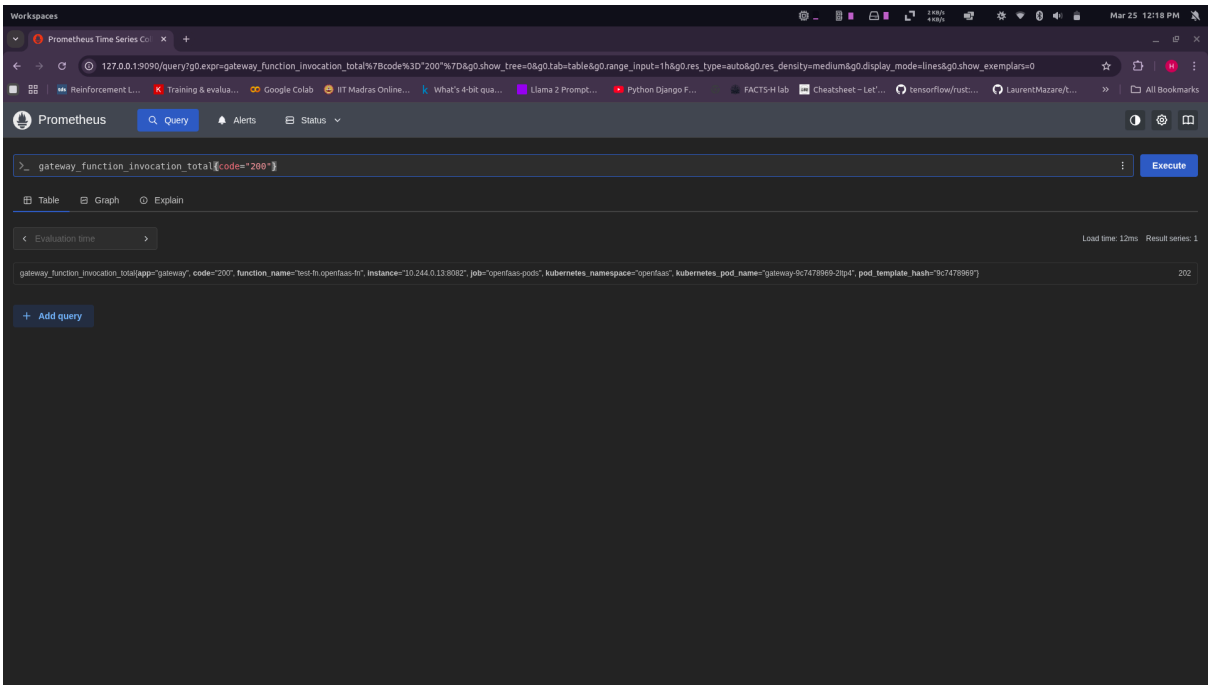
test.py

```
import requests

for i in range(1, 200):
    requests.get("http://127.0.0.1:8080/function/test-fn")
```

Function Latency



Success rate



2)
Terraform

[main.tf]

```
variable "gateway_url" {}
variable "username" {}
variable "password" {}
variable "action" {}

resource "null_resource" "deploy" {
  count = var.action == "deploy" ? 1 : 0

  provisioner "local-exec" {
    command = <<EOT
echo "${var.password}" | faas-cli login \
  --username ${var.username} \
  --password-stdin \
  --gateway ${var.gateway_url}

faas-cli deploy -f ${path.root}/../stack.yml --gateway ${var.gateway_url}
EOT
  }
}

resource "null_resource" "destroy" {
  count = var.action == "destroy" ? 1 : 0

  provisioner "local-exec" {
```

```
  command = <<EOT
echo "${var.password}" | faas-cli login \
  --username ${var.username} \
  --password-stdin \
  --gateway ${var.gateway_url}

faas-cli remove test-fn --gateway ${var.gateway_url}
EOT
  }
}
```

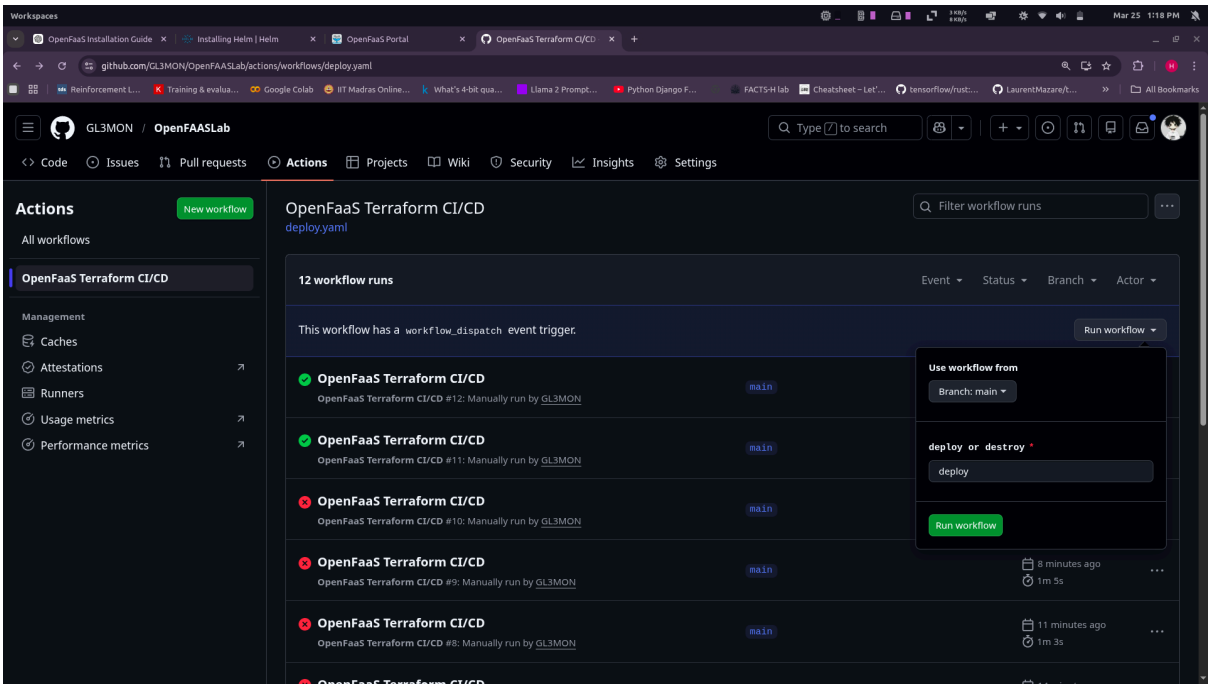
Github workflows

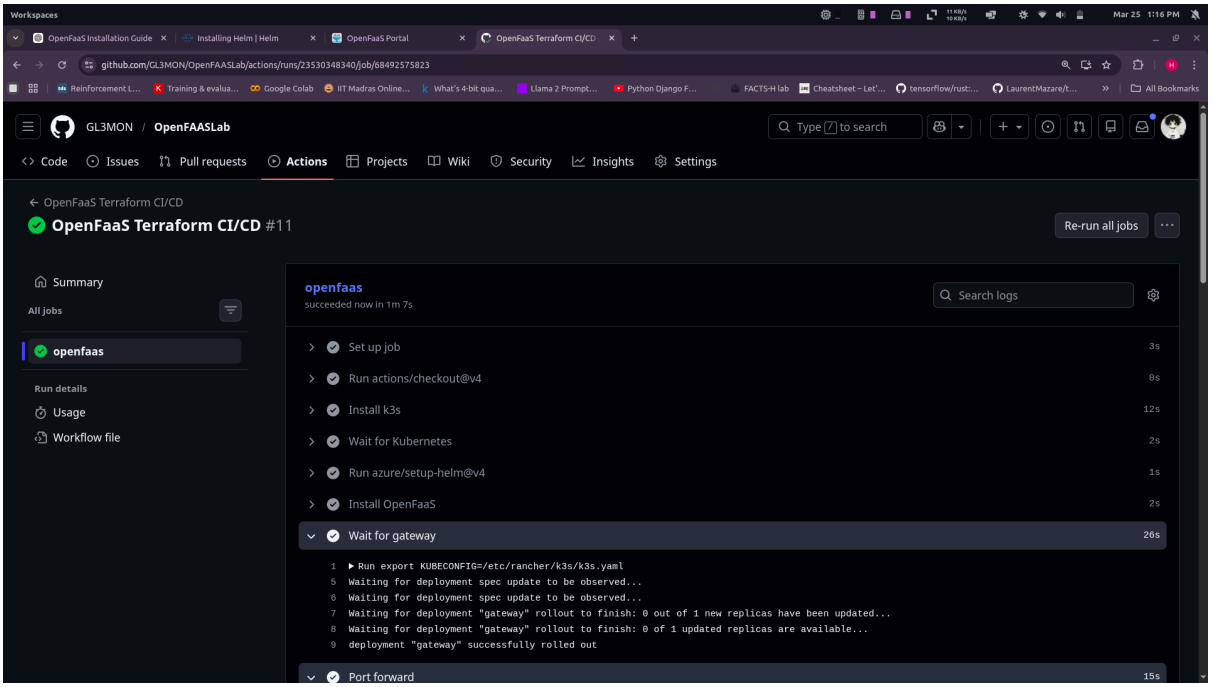
```yaml
name: OpenFaaS Terraform CI/CD

on:
 workflow_dispatch:
   inputs:
     action:
       description: deploy or destroy
       required: true
       default: deploy

jobs:
 openfaas:
   runs-on: ubuntu-latest

   steps:
   - uses: actions/checkout@v4

   # Install k3s
   - name: Install k3s
     run: |
       curl -sfL https://get.k3s.io | sh -
       sudo chmod 644 /etc/rancher/k3s/k3s.yaml

   # Wait for cluster
   - name: Wait for Kubernetes
     run: |
       export KUBECONFIG=/etc/rancher/k3s/k3s.yaml
       kubectl wait --for=condition=Ready node --all --timeout=180s
       kubectl get nodes

   # Install Helm
```

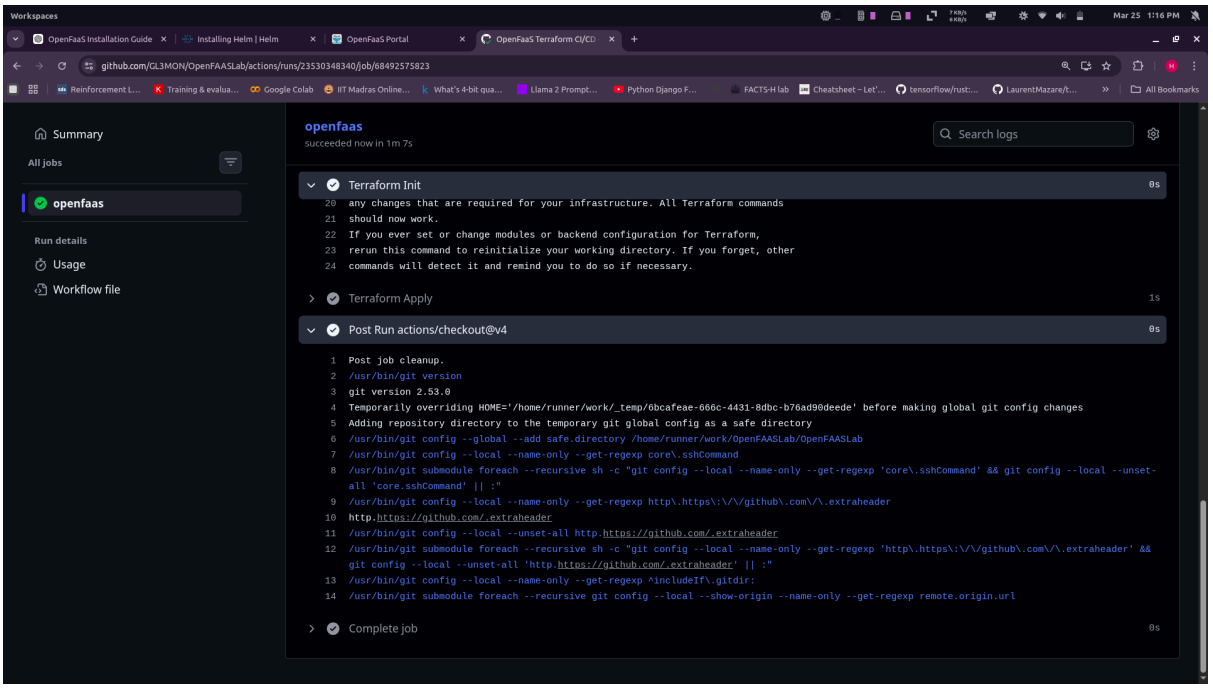
```
    - uses: azure/setup-helm@v4

    # Install OpenFaaS
    - name: Install OpenFaaS
      run: |
        export KUBECONFIG=/etc/rancher/k3s/k3s.yaml

        kubectl apply -f
https://raw.githubusercontent.com/openfaas/faas-netes/master/namespaces
.yml

        helm repo add openfaas https://openfaas.github.io/faas-netes
        helm repo update

        kubectl -n openfaas create secret generic basic-auth \
          --from-literal=basic-auth-user=admin \
          --from-literal=basic-auth-password=password || true

        helm upgrade openfaas openfaas/openfaas \
          --install \
          --namespace openfaas \
          --set functionNamespace=openfaas-fn

    # Wait for gateway
    - name: Wait for OpenFaaS gateway
      run: |
        export KUBECONFIG=/etc/rancher/k3s/k3s.yaml
        kubectl rollout status deploy/gateway -n openfaas --timeout=180s

    # Port forward
    - name: Port forward gateway
      run: |
        export KUBECONFIG=/etc/rancher/k3s/k3s.yaml
        kubectl port-forward svc/gateway -n openfaas 8080:8080 &
        sleep 10

    # Get real password from cluster
    - name: Get OpenFaaS password
      run: |
        export KUBECONFIG=/etc/rancher/k3s/k3s.yaml
        echo "OPENFAAS_PASSWORD=$(kubectl -n openfaas get secret
basic-auth \
```

```
      -o jsonpath='{.data.basic-auth-password}' | base64 --decode)" >>
$GITHUB_ENV

  # Install faas-cli
  - name: Install faas-cli
    run: curl -sSL https://cli.openfaas.com | sudo sh

  # Install Terraform
  - uses: hashicorp/setup-terraform@v3

  # Terraform init
  - name: Terraform Init
    run: |
      cd terraform
      terraform init

  # Terraform apply (deploy OR destroy)
  - name: Terraform Apply
    run: |
      cd terraform
      terraform apply -auto-approve \
        -var gateway_url=http://127.0.0.1:8080 \
        -var username=admin \
        -var password=$OPENFAAS_PASSWORD \
        -var action=${{ github.event.inputs.action }}
```

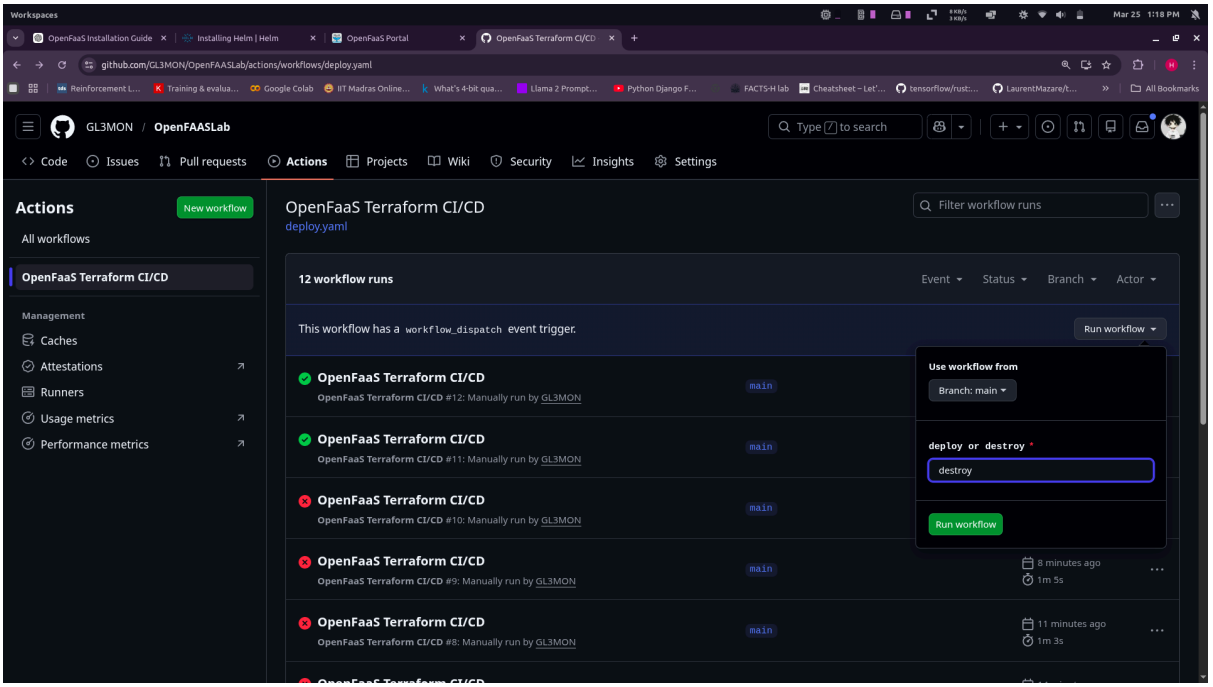
Deploy action







Destroy action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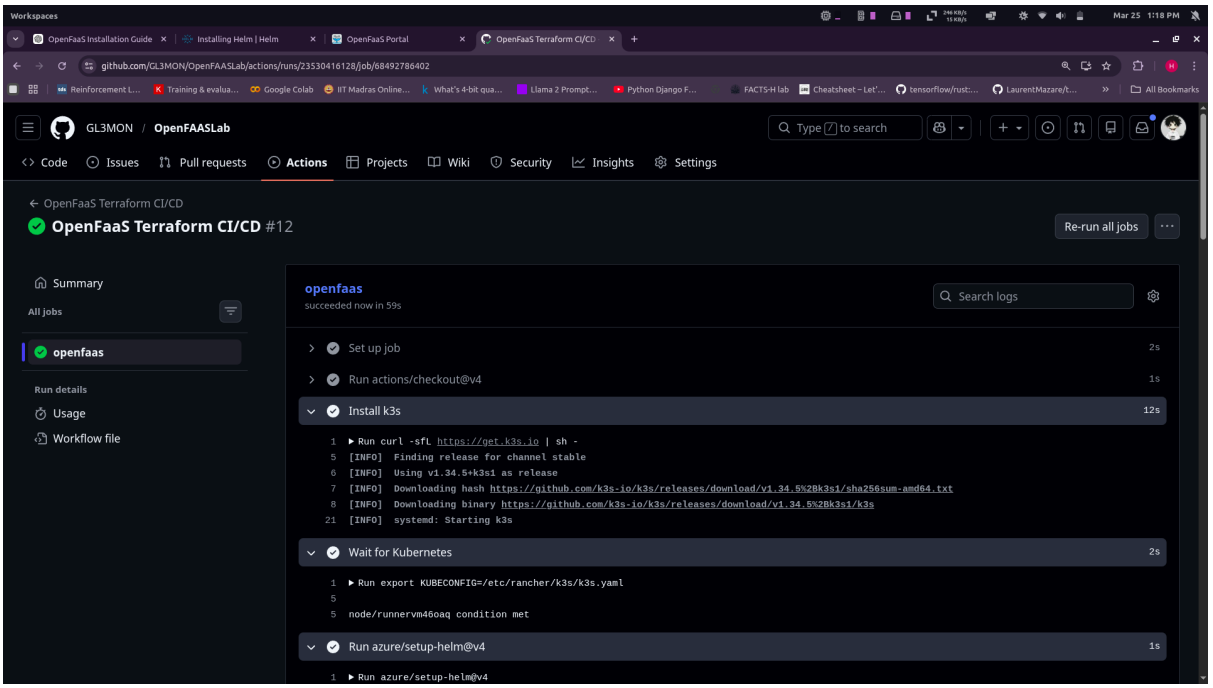
main
8 minutes ago
1m 5s
OpenFaaS Terraform CI/CD
OpenFaaS Terraform CI/CD #8: Manually run by GL3MON
main
11 minutes ago
1m 3s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succeeded now in 59s
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Set up job
2s
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Install k3s
12s
1 ▶ Run curl -sfL https://get.k3s.io | sh -
5 [INFO] Finding release for channel stable
6 [INFO] Using v1.34.5+k3s1 as release
7 [INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.34.5%2Bk3s1/sha256sum-amd64.txt
8 [INFO] Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.34.5%2Bk3s1/k3s
21 [INFO] systemd: Starting k3s
Wait for Kubernetes
2s
1 ▶ Run export KUBECONFIG=/etc/rancher/k3s/k3s.yaml
5
5 node/runnervm46oaq condition met
Run azure/setup-helm@v4
1s
1 ▶ Run azure/setup-helm@v4

Workspaces

OpenFaaS Installation Guide | Installing Helm | Helm | OpenFaaS Portal | OpenFaaS Terraform CI/CD

github.com/GL3MON/OpenFAASLab/actions/runs/23530416128/job/68492786402

Summary

All jobs

openfaas

Run details

Usage

Workflow file

openfaas

succeeded now in 59s

Search logs

Terraform Init 2s

```
any changes that are required for your infrastructure. All Terraform commands
should now work.
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

Terraform Apply 0s

Post Run actions/checkout@v4 0s

```
Post job cleanup.
/usr/bin/git version
git version 2.53.0
Temporarily overriding HOME='/home/runner/work/_temp/10219907-7af0-497a-ab3c-bb60a2c19d47' before making global git config changes
Adding repository directory to the temporary git global config as a safe directory
/usr/bin/git config --global --add safe.directory /home/runner/work/OpenFAASLab/OpenFAASLab
/usr/bin/git config --local --name-only --get-regexp core\.sshCommand
/usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'core\.sshCommand' && git config --local --unset-all 'core.sshCommand' || :"
/usr/bin/git config --local --name-only --get-regexp http\.https\:\/\/github\.com\/\.extraheader
http.https://github.com/.extraheader
/usr/bin/git config --local --unset-all http.https://github.com/.extraheader
/usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'http\.https\:\/\/github\.com\/\.extraheader' && git config --local --unset-all 'http.https://github.com/.extraheader' || :"
/usr/bin/git config --local --name-only --get-regexp ^includeIf\.gitdir:
/usr/bin/git submodule foreach --recursive git config --local --show-origin --name-only --get-regexp remote.origin.url
```

Complete job 0s